

Speech-N-LlaMA: Improving Speech LLMs with Multi-Pass Training

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Abstract—Speech LLMs use speech embeddings as the prompt to a Large Language Model (LLM) and generate human readable text for the speech signal in an autoregressive manner. Teacher-forcing is a common approach used for training Speech LLMs, which is dissimilar to the procedure used during inference, creating a gap between training and inference known as exposure bias. To mitigate exposure bias, we propose Speech-N-LlaMA. Contrary to the existing Speech LLMs that have single pass through the LLM during training, Speech-N-LlaMA incorporates multi-pass training. Through multiple passes, Speech-N-LlaMA mitigates exposure bias and uses the error correction capability of LLM to improve the performance of Speech LLMs. We propose an N-pass loss and utterance level temperature sampling in Speech-N-LlaMA to achieve this. We evaluate four different model sizes on three benchmarks, and show up to 18% relative improvement in Word Error Rate (WER) compared to baseline Speech LLM, while not incurring any additional compute during inference.

Index Terms—Automatic Speech Recognition, Large Language Models, LLaMA, Speech LLMs

I. INTRODUCTION

Conventional Automatic Speech Recognition (ASR) methods include end-to-end approaches such as Attention-based Encoder Decoder (AED) architecture [1], Connectionist Temporal Classification (CTC) [2], and Recurrent Neural Network-Transducer (RNN-T) [3]–[5]. Recent works have used Large Language Models (LLMs) in speech recognition tasks (hereafter, referred as “Speech LLMs”) and have shown promising performance [6]–[8]. In this work, our goal is to improve the performance of an existing Speech LLM.

Speech LLMs and many promising conventional ASR methods such as RNN-T are auto-regressive *i.e.*, during inference, tokens are predicted sequentially, where each token prediction is conditioned on all previously predicted tokens. A common approach used for training auto-regressive methods is sequence-to-sequence learning with substituting previous output token with the ground-truth tokens, which is commonly referred to as teacher-forcing [9]. Teacher-forcing is dissimilar to the procedure used at inference time, when the gold target sequence (*i.e.*, the reference text) is not available and decisions are conditioned on previous model predictions. This creates a gap between training and inference known as exposure bias [10].

Several works have been proposed to mitigate exposure bias in conventional encoder-only [11] and encoder-decoder [12], [13] ASR methods. However, limited work explores reducing exposure bias in Speech LLMs *i.e.*, a decoder-only architecture, which we address in this work. We focus on a particular architecture for Speech LLMs, dubbed Speech-LlaMA [7], [8]. Fig. 1 highlights our main contributions. As shown in Fig. 1, Speech-LlaMA has one pass during training. By “one pass”, we refer to a single iteration of predicting (*i.e.*, decoding) all the tokens in an utterance. Our proposed method, which we name Speech-N-LlaMA, has multiple passes through LLM

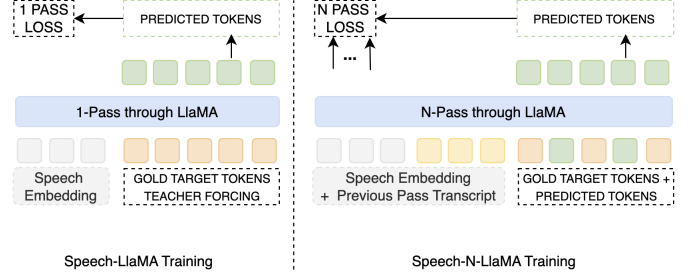


Fig. 1. Block diagram of training SpeechLlaMA (single-pass, teacher forcing), and Speech-N-LlaMA (N-passes, no teacher forcing)

during training. Multiple passes help us in augmenting Speech-LlaMA with two novel additions to get Speech-N-LlaMA.

First, it enables an approach to use schedule sampling [12]–[14] and mitigate exposure bias in Speech LLMs. Speech LLMs have a decoder-only architecture. Some work proposed schedule sampling for decoder only architecture [14], [15]. But the application is on a different task than ASR and, such work have only 2-passes and use 1-pass loss for training which is based on only the outputs from the second decoder pass as shown in Fig. 2. On the contrary, Speech-N-LlaMA has N-passes and utilizes the proposed N-pass loss which uses outputs from all the decoder passes for training as shown in Fig. 2.

Our second addition exposes the Speech LLMs to potential errors in transcriptions and train it to directly avoid them. As shown in Fig. 1, Speech-LlaMA has only speech embedding in the prompt [7], [8], but Speech-N-LlaMA uses both speech embedding and previous-pass predicted transcript as the prompt. This addition is similar to existing work on error correction [16], [17] and deliberation [18] as shown in Fig. 3. In error correction [16], Large Language Model (LLM) like Chat-GPT [19] and LlaMA [20] are

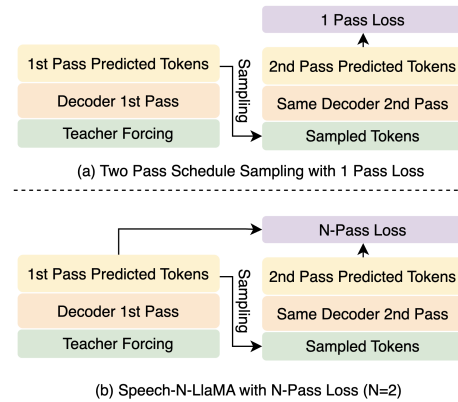


Fig. 2. Difference between schedule sampling and Speech-N-LlaMA.

*Work done during an internship at Meta.

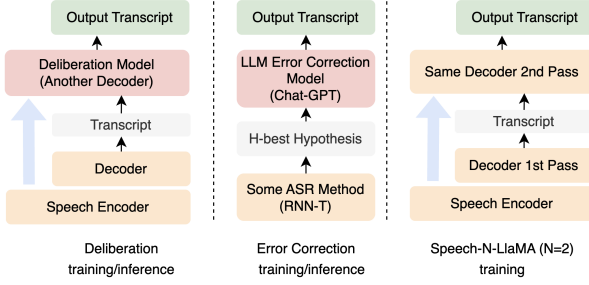


Fig. 3. Difference between deliberation, error correction and Speech-N-Llama for $N = 2$. Contrary to Speech-N-Llama, the former two approaches need an additional deliberation or error correction model.

prompted with H-best hypothesis to generate new transcript. While in deliberation, speech embedding and 1st-pass transcript from a RNN-T goes through a 2nd-pass using a separate deliberation model that correct errors in 1st pass transcript [18]. We differ from both by not needing a separate deliberation model or LLM model to correct errors. Speech-N-Llama uses the same decoding model and performs N -passes through it during training. Moreover both deliberation and LLM based error correction add significant compute during inference, as both of them require another pass with either a deliberation model or an LLM like Chat-GPT. However, our experiments show that our proposed training method directly improves the 1st pass results, without the need of running multi-passes in inference, thus not incurring any additional compute cost during inference.

II. RELATED WORK

Several works examine Speech LLM for different speech recognition tasks [6]–[8], [21]. Wu *et al.* [7] and Fathullah *et al.* [8] used LLaMA-7B, and proposed a decoder-only architecture for multilingual speech translation. Wu *et al.* [7] showed that using a pre-trained CTC based speech encoder to obtain speech features results in better performance. Fathullah *et al.* [8] used a conformer based speech encoder trained using CTC loss to obtain speech features. Following these findings, in our Speech-Llama for ASR, we also use CTC based speech encoder. Some work also examined Whisper [22] to obtain speech features and Vicuna [23] LLM for ASR [21]. In all of these works, Speech LLMs are trained either from scratch or using Low-rank Adaptation (LoRA)[24]. During training, all of these works use sequence-to-sequence learning [9] and do a single pass through LLM using teacher forcing, which results in exposure bias [10]. On the contrary, in this work, we do multiple passes and use it to mitigate exposure bias in Speech LLMs.

Some work has been done on mitigating exposure bias in encoder-only [11] and encoder-decoder ASR methods [12], [13]. Scheduled sampling [12] is a common approach used to mitigate exposure bias. Very limited work explores this for decoder-only ASR methods such as Speech LLMs [8], [21]. Mihaylova *et al.* [14] proposed an approach which used two-pass decoding to implement scheduled sampling for transformers. However, they used it for machine translation tasks and use 1-pass loss for training which is based on only the outputs from the second decoder pass. In this work, we propose schedule sampling for decoder-only ASR methods. We use outputs from all the decoder passes to calculate our proposed N -pass loss and use transcript from previous pass in the prompt when decoding.

Adding transcript from previous pass in the prompt helps to expose LLM with potential errors in transcription and remove it in next decoder pass. This is inspired by the error correction ability of LLMs as shown in previous work [16]. While previous work [16], [17] post

process transcript generated using a separate LLM to correct errors in the transcript, Speech-N-Llama uses the transcript from previous pass and uses the same LLM to do another pass and corrects error in the transcript. This is similar to what is done in deliberation [18] however in deliberation a separate model is used to correct and do 2nd pass. While we use the same LLM to do multi-passes in training. Also, both of these existing approaches add significant compute during inference. Our experiments show that our proposed training method directly improves the 1st pass results, without the need of running multi-passes in inference.

III. METHOD

Speech LLMs such as Speech-Llama have shown promising ASR performance [8], [21]. In this work, our goal is to improve upon the performance of Speech-Llama, by doing multiple passes through the LLM during training. We introduce an N -pass loss that trains Speech-N-Llama to correct its own generated hypothesis in previous passes and mitigate exposure bias. Below we first briefly discuss CTC ASR and Speech-Llama, then describe our proposed Speech-N-Llama.

A. CTC ASR and Speech-Llama

We use two main baselines in our work: CTC ASR and Speech-Llama. Fig. 4 shows a block diagram for both methods. The CTC ASR converts a speech signal to a mel-filter bank [25] and processes it using a series of VGG [26] and Conformer blocks [1], and the model is trained using a CTC loss. The Speech-Llama model is constructed as in [7], [8]. The output of the last conformer layer from the CTC encoder is used, which amounts to a sequence of speech embeddings $\mathbf{h}$. The latter is fed to an LLM decoder as prompt. Following the prompt, the LLM decoder is trained for ASR using next token prediction with teacher forcing [9]. The LLM decoder is initialized from a pretrained 125M parameters model, that was pretrained in-house for next-token prediction using a large text corpus. The CTC encoder and the LLM decoder are finetuned jointly on the ASR task.

B. First Pass in Speech-N-Llama

Our proposed method extends the Speech-Llama model to include multiple decoding passes during training, using a shared LLM decoder. In the first pass, Speech-N-Llama uses teacher forcing [9], *i.e.*, processes $\mathbf{Y}_t = \{< bos >, \mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{L-1}\}$ as input and is trained to emit the ground truth $\mathbf{Y}_{ref} = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_L\}$, where L is the length of the reference token sequence including the $< eos >$ symbol. The LLM decoder output is denoted by $\mathbf{Y}_o = \{\mathbf{o}_1, \mathbf{o}_2, \dots, \mathbf{o}_L\}$, each contains logits for all tokens in the vocabulary V . The first pass loss then given by:

$$Loss_1 = - \sum_{i=1}^L \log \text{softmax}(\mathbf{o}_i)_{\mathbf{y}_i} \quad (1)$$

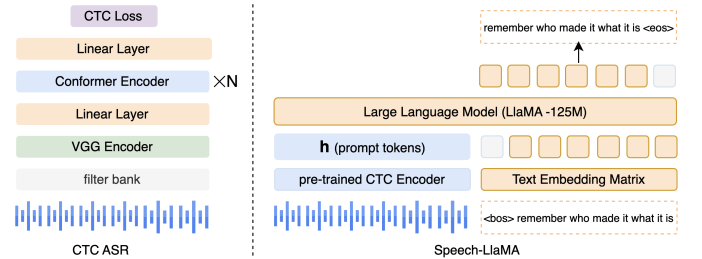


Fig. 4. Block diagram of baseline method Speech-Llama. A pretrained CTC model provides speech embeddings that are in turn fed as a prompt into a Llama decoder which is trained for ASR.

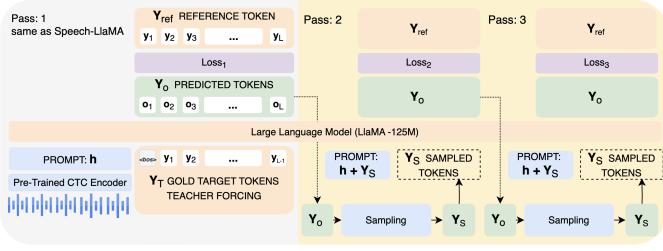


Fig. 5. Training Speech-N-LLaMA, $N = 3$.

C. Next Passes in Speech-N-LLaMA

For the second pass, Speech-N-LLaMA replaces the reference text tokens as input to the model with a sample from the output distribution of the first pass. To control the diversity in the sampling, temperature scaling is applied over the output Y_o . A new token sequence Y_S is then sampled from the distribution Y_o after temperature scaling. As we sample the complete sequence Y_S together, we refer to it as utterance level temperature sampling. In order to mitigate exposure bias, the sequence Y_S is used as input to the model instead of performing teacher forcing using Y_T .

Similarly, we generate sampled tokens for a third pass using tokens predicted in the second pass. Note that all passes use the shared LLM decoder, therefore minimizing exposure bias by requiring it to generate the correct token sequence even when previous tokens included errors. The loss term $Loss_k$ associated with pass k is identical to the one from the first pass in Eq. (1), when the Y_o is now the LLM decoder output from pass k . Some works show that LLMs have the ability to correct errors in transcript [16]. Inspired by this, we append the transcript obtained from pass $k - 1$ to the LLM decoder prompt in pass k , where $k \in \{2, \dots, N\}$. This gives the LLM a full context lookahead into the transcript obtained by previous passes, aiding its error correction ability.

Speech-LLaMA and existing work which use two-pass decoding to mitigate exposure bias in transformers [14], have only a single loss term. In contrast, in Speech-N-LLaMA we associate each pass with a loss function, and the final N-pass loss is a weighted sum of the different passes losses:

$$N\text{-pass Loss} = \sum_{k=1}^N w_k Loss_k. \quad (2)$$

Note that training the model to emit the reference text in multiple passes is hypothesized to improve exposure bias. During inference however as shown in Fig. 6, we perform one pass only using the trained model, therefore not incurring any additional inference cost compared to the baseline Speech-LLaMA model.

IV. EXPERIMENTS

This section discusses training strategy, dataset, metric and results.

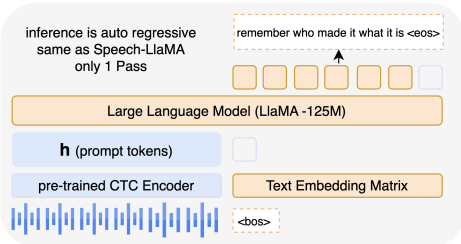


Fig. 6. Inference from Speech-N-LLaMA.

A. Datasets and Evaluation Metric

We use three datasets in our experiments: LibriSpeech (LS) [27], English part of Multilingual LibriSpeech (MLS-En) [28] and TED-LIUM [29]. For training, validation and testing, we use the official splits. For LS, the dev-other split is used for validation and model selection, and test-clean and test-other splits are used for testing. The LS has 960 hours of labeled speech for training while MLS-En and TED-LIUM contains about 10K hours and 452 hours of labeled speech, respectively. We use Word Error Rate (WER) as our evaluation metric. We also report Oracle WER [30], which is the lowest WER across the H hypotheses returned from the H -best beam search inference. We use $H = 4$ to report Oracle WER in all our experiments.

B. Models and Training Details

We use 80-dimensional log mel-filterbanks [25] computed across 25ms with a 10ms frame stride, to obtain speech features. In our CTC ASR, each Conformer layer has an input dimension of 768 and uses 8 attention heads. Four different sizes of CTC ASR were experimented with by varying the number of conformer layers. We have 3, 6, 12, and 24 conformer encoder layers in 25M, 50M, 100M, and 200M parameters CTC ASR, respectively. Four different sizes of the CTC encoder lead to four different sizes of the Speech-LLaMA model. We purposely keep the size of the LLM fixed in all different sizes of Speech-LLaMA because doubling the size of the speech encoder reduces WER more than doubling the size of the LLM [21]. We remove the last linear layer from CTC ASR, and use it in Speech-LLaMA and Speech-N-LLaMA, as a pre-trained CTC encoder. We stack features from 3 consecutive frames and using a linear layer, project it to the input dimension of the LLM. The LLM decoder contained 12 layers from the LLaMA architecture [20], [31], each with 12 attention heads and an input dimension of 768, and is initialized from a 125M parameters model pretrained in-house for next-token prediction using a large text corpus. The Adam [32] optimizer is used for training. Unless otherwise mentioned, our proposed Speech-N-LLaMA is using 3 passes during training.

TABLE I
COMPARISON OF PROPOSED SPEECH-N-LLAMA WITH CTC AND SPEECH-LLAMA ON LIBRISPEECH, MLS-EN AND TED-LIUM DATASETS. WE REPORT WER (IN %) OF DIFFERENT ENCODER (ENC.) SIZE AND THE ΔWER IMPROVEMENT BY SPEECH-N-LLAMA RELATIVE TO SPEECH-LLAMA.

Enc. Size	Method	LibriSpeech			MLS-En		TED-LIUM	
		test clean	test other	oracle WER	test	oracle WER	test	oracle WER
25M	CTC ASR	5.56	14.04	11.27	15.60	14.02	11.80	9.42
	Speech-LLaMA	4.27	8.49	6.13	7.77	6.24	10.01	7.71
	Speech-N-LLaMA	3.74	7.65	5.20	7.65	5.85	9.38	7.33
	$\Delta WER \%$	$\downarrow 12.41$	$\downarrow 9.90$	$\downarrow 15.17$	$\downarrow 1.54$	$\downarrow 6.25$	$\downarrow 6.29$	$\downarrow 4.93$
50M	CTC ASR	4.02	10.05	7.71	10.73	9.27	9.16	7.15
	Speech-LLaMA	3.86	7.21	5.16	6.74	5.28	8.88	6.87
	Speech-N-LLaMA	3.25	6.70	4.58	6.59	4.93	8.27	6.66
	$\Delta WER \%$	$\downarrow 15.80$	$\downarrow 7.07$	$\downarrow 11.24$	$\downarrow 2.23$	$\downarrow 6.63$	$\downarrow 6.87$	$\downarrow 3.06$
100M	CTC ASR	3.60	9.06	6.77	8.73	7.35	8.49	6.53
	Speech-LLaMA	3.92	6.94	4.96	6.07	4.69	8.19	6.47
	Speech-N-LLaMA	3.20	6.41	4.37	5.94	4.42	8.07	6.33
	$\Delta WER \%$	$\downarrow 18.37$	$\downarrow 7.64$	$\downarrow 11.90$	$\downarrow 2.14$	$\downarrow 5.76$	$\downarrow 1.47$	$\downarrow 2.16$
200M	CTC ASR	3.45	8.56	6.48	8.12	6.80	8.10	6.27
	Speech-LLaMA	3.57	7.08	5.12	6.01	4.66	7.12	5.49
	Speech-N-LLaMA	3.37	6.31	4.50	5.71	4.23	6.92	5.32
	$\Delta WER \%$	$\downarrow 5.60$	$\downarrow 10.88$	$\downarrow 12.11$	$\downarrow 4.99$	$\downarrow 9.23$	$\downarrow 2.81$	$\downarrow 3.10$
Avg. $\Delta WER \%$		$\downarrow 13.05$	$\downarrow 8.87$	$\downarrow 12.61$	$\downarrow 2.73$	$\downarrow 6.97$	$\downarrow 4.36$	$\downarrow 3.31$

C. Main Results

We evaluate Speech-N-LLaMA and compare its performance with the baseline methods Speech-LLaMA and CTC. First, note that our baseline implementation of Speech-LLaMA is competitive compared to Speech LLMs proposed in existing work. For instance, Yu *et al.*'s Whisper-base(74M)+Vicuna(13B) Speech LLM [21] achieves a WER of 9.83% on the LS test-other subset while our CTC(25M)+LLaMA(125M) achieves a WER of 8.49%. In Table I, we report WER on the LS, MLS-En, and TED-LIUM datasets. For LS, the reported Oracle WER is on the test-other subset. In Table I, we observe that as expected, for each approach, a larger network size results in better performance. Comparing the performance of CTC and Speech-LLaMA, we observe that adding an LLM to the CTC encoder leads to a considerable improvement in performance. An interesting observation is that adding 125M LLM to 25M CTC encoder *i.e.*, 150M Speech LLM has similar or better performance as 200M CTC ASR. For all model sizes and subsets, our proposed Speech-N-LLaMA performs better than Speech-LLaMA, with a relative improvement in WER of 1%-18%. In addition, Speech-N-LLaMA improves oracle WER *i.e.*, hypothesis quality by 2%-15% relative for different model sizes compared to the Speech-LLaMA model.

D. Comparison with Existing Approaches

Speech-N-LLaMA mitigate exposure bias and use error correction capability of LLMs. Prabhavalkar *et al.* [33] proposed Minimum Word Error Rate (MinWER) training while Mihaylova *et al.* [14] proposed 2-pass decoding to mitigate exposure bias. We implemented them for Speech-LLaMA, and compared its performance with Speech-N-LLaMA. Our proposed Speech-N-LLaMA has better performance than both of these approaches. Since our proposed method may be viewed as correcting errors in transcription obtained from previous passes, we also provide the performance that can be obtained using larger text-based LLMs for error correction. We implemented two approaches: one-shot-unconstrained and one-shot-closest as given in [16]. Given H-hypothesis, the one-shot-unconstrained approach uses one-shot learning and uses LLM for unconstrained generation of new transcript while one-shot-closest approach uses the unconstrained generated transcript to find the closest match in the given H-hypothesis. We use 4-best hypotheses generated from Speech-LLaMA and use LLaMA-3-70B [31] and LLaMA-3.1-405B [20] for error correction. LLaMA-3.1-405B [20] with just one example of error-correction can help in correcting error in transcript. All results are reported in Table II. Speech-N-LLaMA performs better than all the compared methods.

TABLE II
COMPARISON OF SPEECH-N-LLAMA WITH MINWER AND 2-PASS SCHEDULE SAMPLING TO MITIGATE EXPOSURE BIAS AND LLM BASED ERROR CORRECTION. WE REPORT WER (IN %) ON LIBRISPEECH DATASET.

Encoder Size	Model	test-clean	test-other
25M	Speech-LLaMA	4.27	8.49
25M	MinWER	4.46	8.65
25M	2-pass schedule sampling	4.03	7.96
25M	Using LLaMA-3.0-70B one-shot-unconstrained	4.97	9.30
25M	one-shot-closest	4.62	8.89
25M	Using LLaMA-3.1-405B one-shot-unconstrained	4.00	8.12
25M	one-shot-closest	3.99	8.05
25M	Speech-N-LLaMA	3.74	7.65

TABLE III

RESULTS OF ABLATION STUDIES ON LIBRISPEECH DEV-OTHER SUBSET, AS DISCUSSED IN SEC. IV-E. WE REPORT WER (IN %).

Hyperparameters		WER
temperature scaling for schedule sampling	teacher forcing	8.37
	$\alpha = 1.5$	8.32
	$\alpha = 1.0$	8.07
	$\alpha = 0.8$	7.72
prompt error correction	$\alpha = 0.5$	8.20
prompt error correction	$\mathbf{Y}_s$ not in prompt	7.78
	$\mathbf{Y}_s$ in prompt	7.72
loss weight w_2/w_1	0	8.37
	1	7.72
	2	7.65
	3	7.73
number of passes N	4	7.88
	1	8.37
	2	7.72
	3	7.47
	4	7.63

E. Further Ablation Studies

We perform further ablation studies to understand the factors contributing to the success of our proposed method. All the below results refer to Table III.

Experiment 1: Temperature Scaling for Scheduled Sampling. We fix the number of passes to two and compare the second pass with teacher forcing and with different temperature for scheduled sampling as described in Section III-C. Recall that in temperature scaling, the logits are multiplied by a factor α before sampling. Therefore a value of $\alpha < 1$ results in increased sampling diversity. The results in Table III highlight that a temperature of $\alpha = 0.8$ is preferred for mitigating exposure bias.

Experiment 2: Prompt Error Correction. We fix the number of passes to two, and use temperature scheduling with $\alpha = 0.8$. We assess the contribution of adding the transcript from a previous pass to the next pass prompt. We found that using $\mathbf{Y}_s$ in the prompt has a small contribution to improved accuracy in Speech-N-LLaMA.

Experiment 3: Loss Weight. We use the best settings from above ablations for the second pass and study the effect of the ratio between the different passes' loss terms in Eq. 2. We found that using twice the weight for second pass loss w_2 is most effective.

Experiment 4: Number of Passes. Once we determine the ideal parameters for the second pass, we perform another ablation on the number of passes. Here we found that using 3 passes yields further accuracy improvements compare to 2 passes only, but accuracy improvement saturates after that.

V. CONCLUSION AND FUTURE WORK

In this work, our goal was to improve the performance of Speech-LLaMA. We propose Speech-N-LLaMA. Contrary to 1-pass for training in the existing Speech LLM, Speech-N-LLaMA uses N-passes during training. N-passes help to mitigate exposure bias and utilize the error correction capability of LLM. Speech-N-LLaMA uses novel utterance level temperature sampling and N-pass loss. We tested 4 different sizes of Speech-N-LLaMA and evaluated them on multiple datasets. Our results show that Speech-N-LLaMA improves the performance by up to 18% compared to Speech-LLaMA, while not incurring any additional compute during inference. In future, we plan to use lattice instead of sampled tokens for error correction and instead of weight sharing LLM, we plan to examine LoRA for N-pass training.

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